

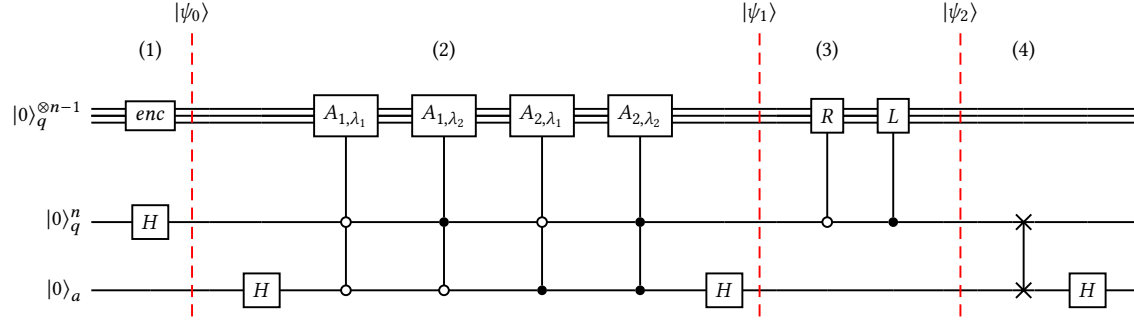
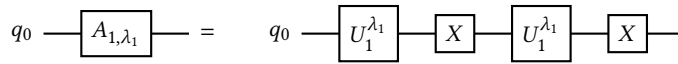


Fig. 3. Quantum circuit to solve the 1D advection-diffusion equation with the lattice Boltzmann method.

Fig. 4. Quantum circuit to implement the A_{1,λ_1} operator.

here. We use the re-encoding of the initial state at each time step as an example of a hybrid design and study the implementation from this particular angle. Note that following this approach likely destroys the possible speed-up advantages of the quantum algorithm. However, it provides an interesting point of study from the program design perspective, which can be useful even after the time-stepping problem for the QLBM has been resolved. Recent attempts towards efficient time-step concatenation can be found, for example, in [Schalkers and Möller 2024a].

The gate complexity of the QLBM depends on the resolution of the computational domain, as detailed in [Budinski 2021, Section 3.3]. It is highlighted that the encoding step is the most resource-intensive, requiring a significant number of state preparation steps and CX gates, which depends on the problem to be solved – in the most general case, this cost is exponential. In the collision step, the complexity is not directly affected by the resolution of the computational domain: increasing the number of qubits to achieve a higher resolution does not change the size of the ancillary register, and the number of multi-qubit gate operations required in the q register is constant. The propagation step is influenced by the resolution of the computational domain, and the depth of this subroutine increases logarithmically. The complexity of the propagation step can be optimized for example by using the parallel state shift method [Budinski et al. 2023]. The overall complexity of the algorithm, excluding the encoding step, is determined to be $O(\log_2(\alpha D))$, where α represents the number of velocity vectors and D depends on the domain resolution ($D = 2M$ for D1Q2).

4 QUANTUM ALGORITHM IMPLEMENTATION IN INTEL QUANTUM SDK

In the form presented above, the quantum lattice Boltzmann algorithm is a hybrid method, although not a variational one. This hybrid property makes it, nevertheless, an interesting example for implementation on platforms that support hybrid classical and quantum computing, such as the Intel Quantum SDK.

In this section, we detail such an implementation and explore the use of modularization with the aim of creating easily reusable and scalable quantum circuits.